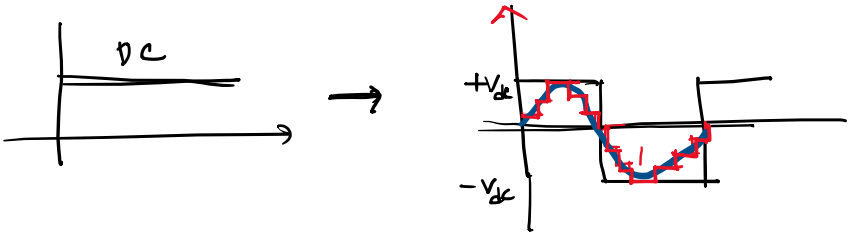
Multi-level Inverter

Friday, May 1, 2026 11:53 AM



$V_{01} = 0.90 V_{dc}$

$V_0 = V_{dc}$

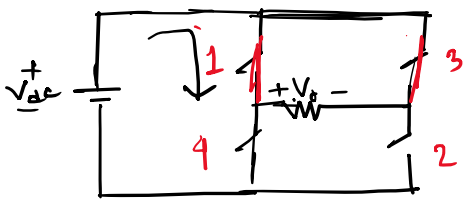
$$THD = \frac{1}{V_{01}} \sqrt{V_0^2 - V_{01}^2}$$

$= 48.34\%$

More sinusoidal-like, Harmonic ↓

THD ↓

##



1,2 ON 3,4 OFF $V_0 = +V_{dc}$

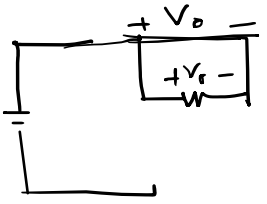
1,2 OFF 3,4 ON $V_0 = -V_{dc}$

1,3 ON $V_0 = 0$

2,4 ON $V_0 = 0$

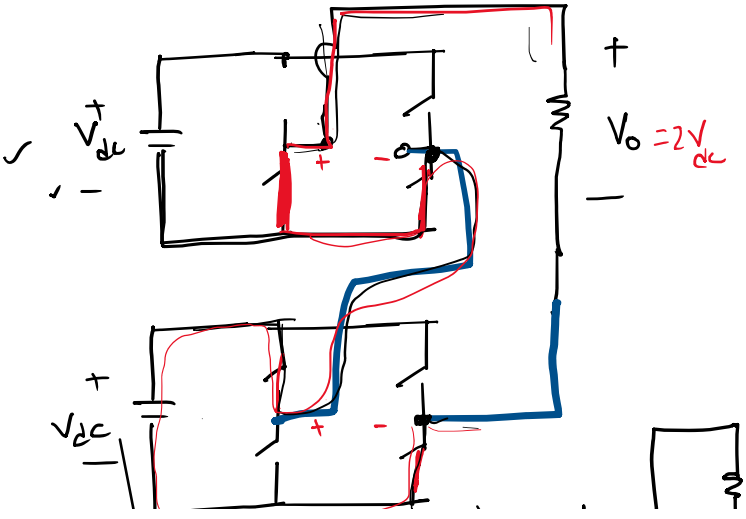
1,4 ✗

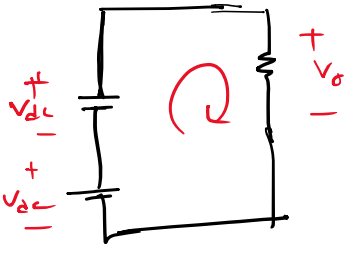
2,3 ✗



$+V_{dc}$, $-V_{dc}$, 0 → 3 level

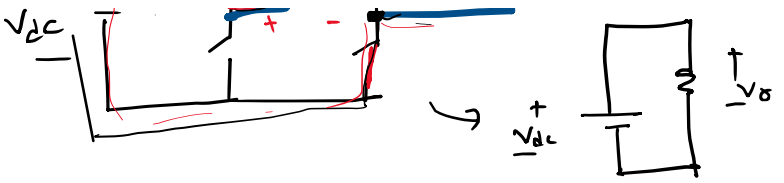
##





$V_0 = +2V_{dc}$

V_{dc}



V_{dc}

$V_o = +V_{dc}$

7-level Inverter

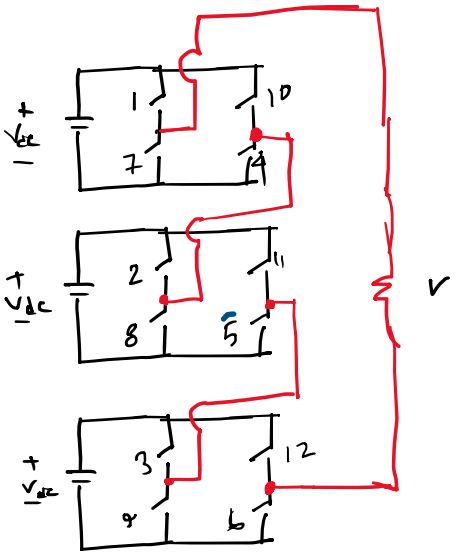
$n=2, \quad 2n+1=5$

$n=3, \quad 2n+1=7$

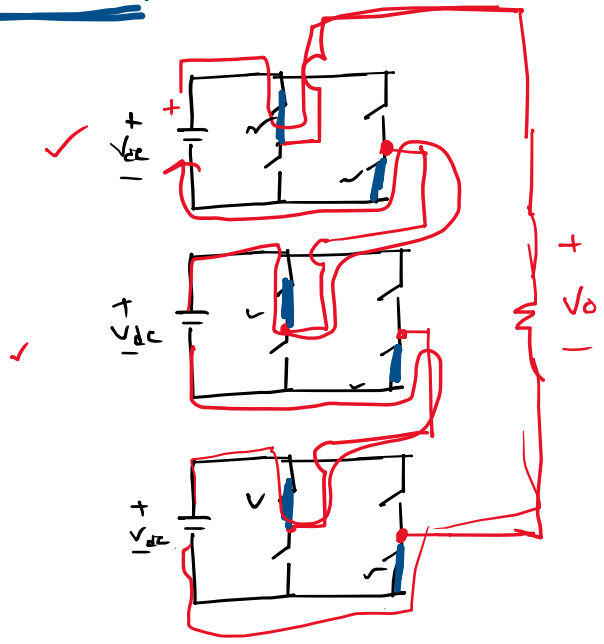
9 level

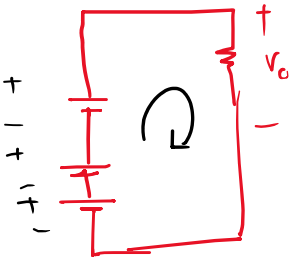
$2n+1 = 9$

$\Rightarrow n=4$



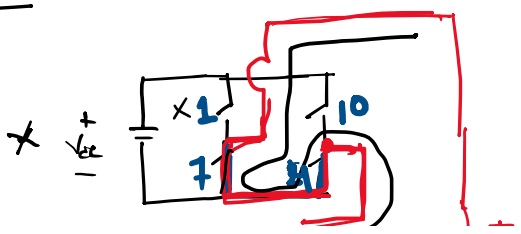
1, 2, 3, 4, 5, 6 :-

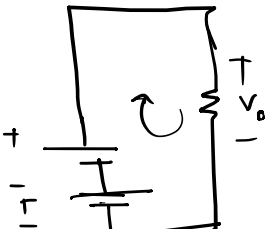


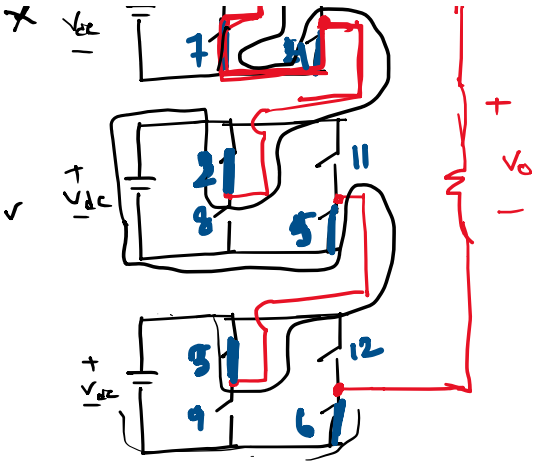


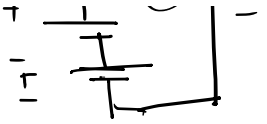
$V_o = +3V_{dc}$

2, 3, 4, 5, 6, 7 :



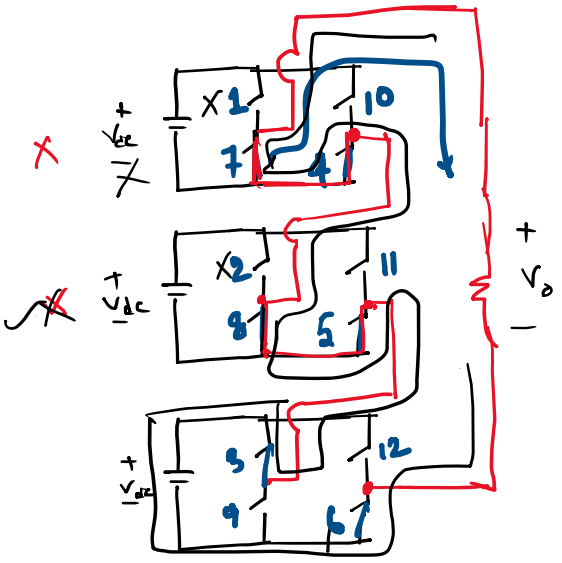


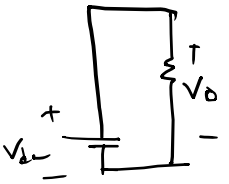




$V_o = +2V_{dc}$

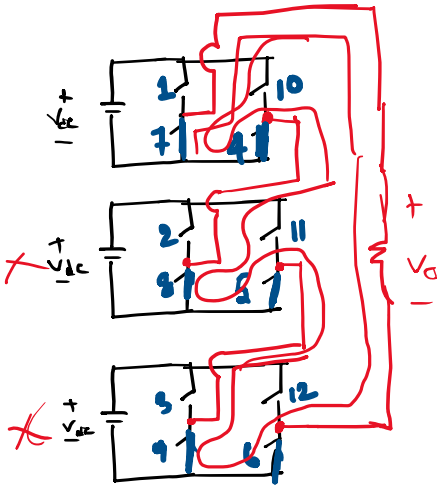
3, 4, 5, 6, 7, 8:

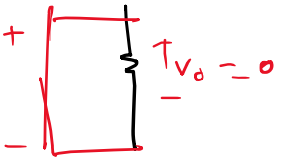




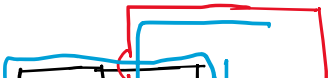
$V_o = +V_{dc}$

4, 5, 6, 7, 8, 9

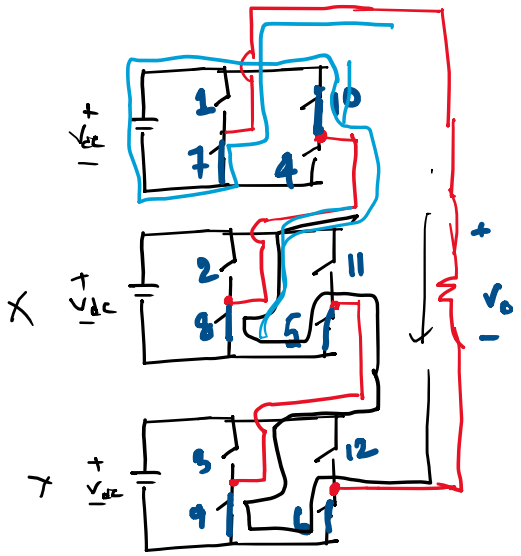


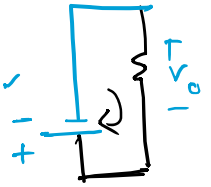


5, 6, 7, 8, 9, 10





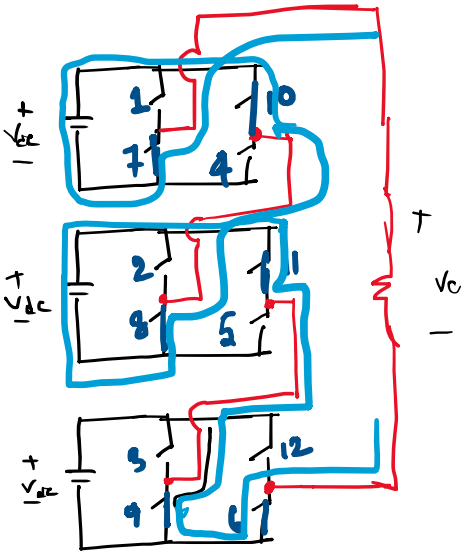


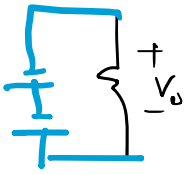


$V_o + V_{dc} = 0$

$\Rightarrow V_o = -V_{dc}$

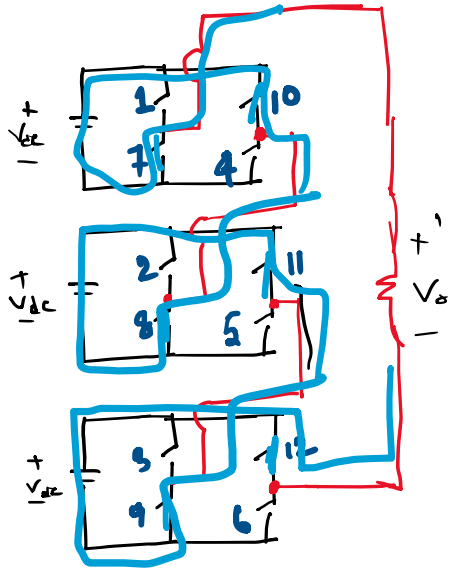
6, 7, 8, 9, 10, 11

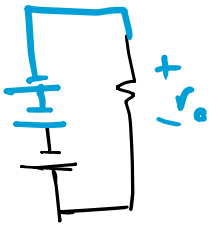




$V_o = -2V_{dc}$

7, 8, 9, 10, 11, 12





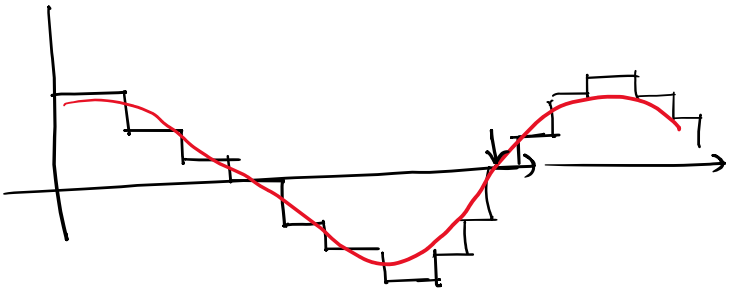
$V_o = -3V_{dc}$

##

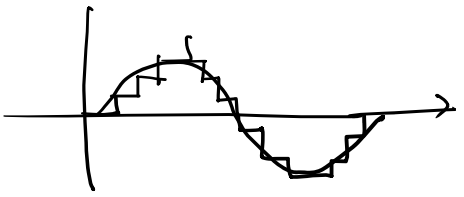
sequence output

##

sequence	output
1,2,3,4,5,6 ⟶	$+3V_{dc}$
2,3,4,5,6,7 ⟶	$+2V_{dc}$
3,4,5,6,7,8 ⟶	$+V_{dc}$
4,5,6,7,8,9 ⟶	0
5,6,7,8,9,10 ⟶	$-V_{dc}$
6,7,8,9,10,11 ⟶	$-2V_{dc}$
7,8,9,10,11,12 ⟶	$-3V_{dc}$

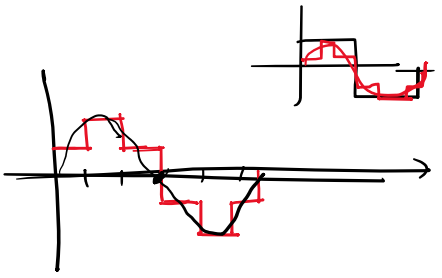


7 level



##

$$V_{o,rms}^2 = \frac{1}{\pi}\left[\int_0^{\pi/3} V_{dc}^2\, d\theta + \int_{\pi/3}^{2\pi/3} (2V_{dc})^2\, d\theta + \int_{2\pi/3}^{\pi} (V_{dc})^2\, d\theta\right]$$



$$= \frac{1}{3}(1+4+1)\,V_{dc}^2$$

0
$\pm V_{dc}$
$\pm 2V_{dc}$

$$\boxed{V_{o,rms} = \sqrt{2}\,V_{dc}} \quad \text{or} \quad (1.41)\,V_{dc}$$

V_{dc}

#

$$V_{01,p} = \frac{6V_{DC}}{\pi}$$

$$V_{01,rms} = \frac{6V_{dc}}{\sqrt{2}\pi} \simeq 1.35\,V_{dc}$$

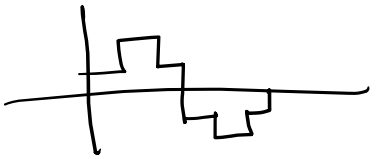
0.90

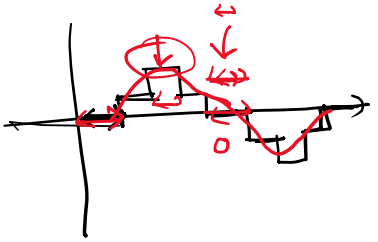
$$THD = \frac{1}{V_{01}}\sqrt{V_0^2 - V_{01}^2} = \frac{1}{1.35}\sqrt{(\sqrt{2})^2 - (1.35)^2}$$

$$= (31.5\%) \leftarrow (48.3\%)$$

level ↑ THD ↓ more sinusoidal

#





THD↓ (20%.)